Digital Design Verification

Lab Manual # 21 – Pipelining

Release: 1.0

Date: 24-June-2024

NUST Chip Design Centre (NCDC), Islamabad, Pakistan

Revision History

Revision Number	Revision Date	Revision By	Nature of Revision	Approved By
1.0	24/06/2024	Ali Aqdas	Complete manual	-
2.0	19/08/2025	Umer Farooq / Fawad Ahmed	Manual Update	

Contents

Objective

The objective of this lab is to

- Pipeline the Design to Enhance the Number Instructions Executed Per Cycle

Tools

- Cadence Xcelium

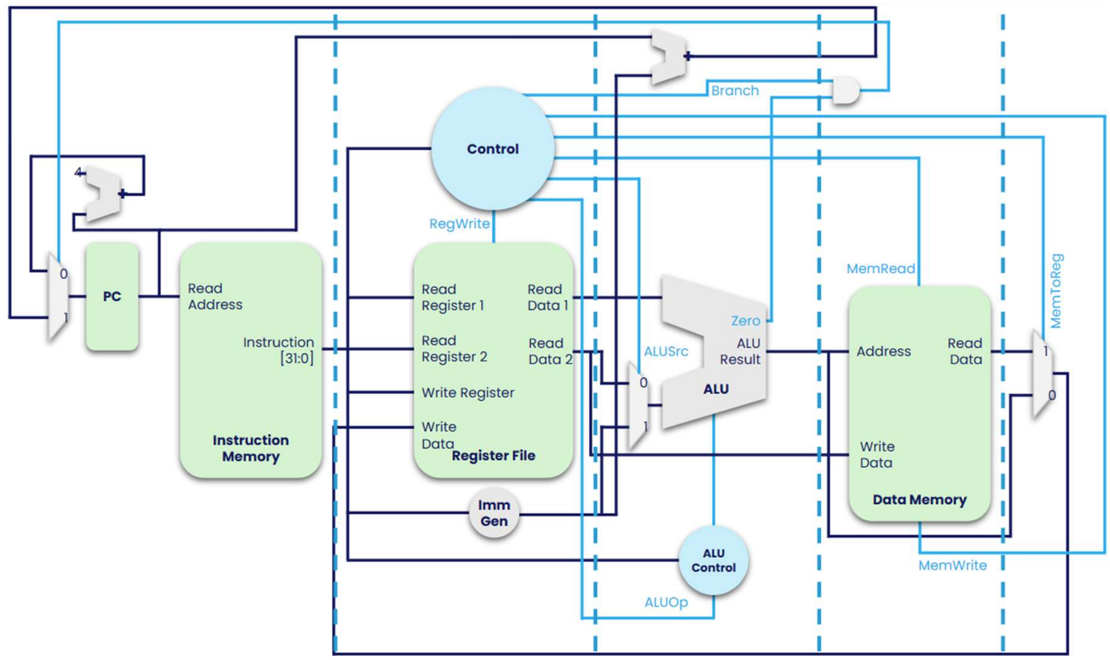
Instructions

The students are required to design and submit the updated files for evaluation of this lab. An additional file for parametric pipeline register has to be added to the submission.

./<student-name>/	./rtl/	program_counter.sv inst_mem.sv data_mem.sv reg_file.sv imm_gen.sv alu_logic.sv branch_comp.sv control_unit.sv IF_ID_Stage.sv ID_EX_Stage.sv EX_MEM_Stage.sv MEM_WB_Stage.sv top.sv
	./tb/	program_counter_tb.sv inst_mem_tb.sv data_mem_tb.sv reg_file_tb.sv imm_gen_tb.sv alu_logic_tb.sv branch_comp_tb.sv control_unit_tb.sv IF_ID_tb.sv ID_EX_tb.sv EX_MEM_tb.sv MEM_WB_tb.sv top_tb.sv

RISC-V Data and Control Paths

The data and control paths designed so far are shown in the figure below.

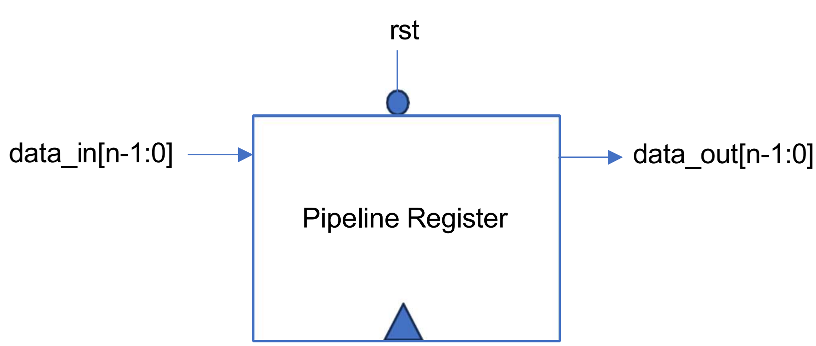


It includes all the **base integer type instructions** and a **control unit** to control multiplexors and writes.

Pipelining: Datapath

A pipelined datapath needs to "separate" the five stages so that each stage can process data from different instruction. To obtain this functionality, intermediate states between different stages must be stored in a state element i.e. a pipeline register.

A pipeline register may have an n-bit input, a reference clock, a reset signal and a n-bit output. A reference block diagram in shown in the figure below.



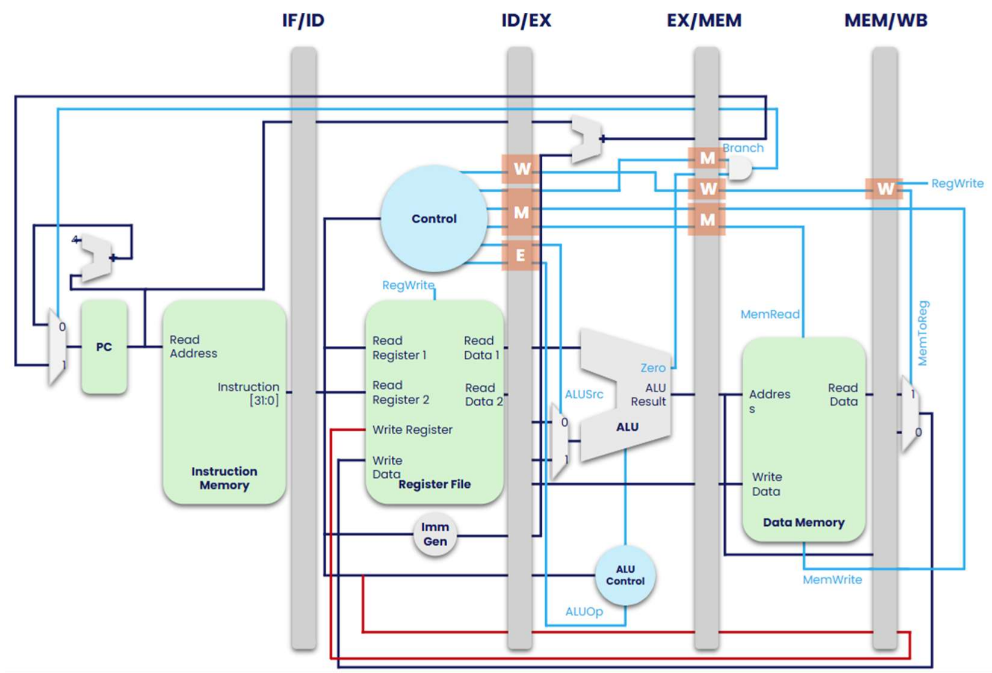
There are two main benefits of pipelining the data and control paths

1. Reduces the critical path, and hence increases the maximum operating frequency.
2. Allows instruction level parallelism, i.e. multiple instructions are being executed in different stages of the processor.

Although these benefits may be at the cost of increased execution time of a single instruction, the true effect of pipelining can be observed in longer executions of entire programs.

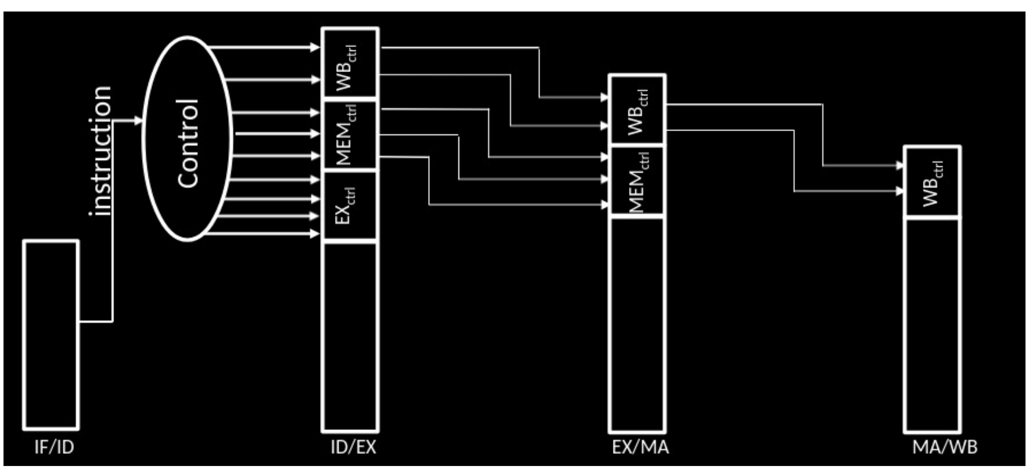
Integration of Pipeline Registers in Datapath

A brief overview of the datapath with pipeline registers can be seen in the following figure. It is a visual depiction of separated stages of the datapath.



Pipeling: Control Path

In addition to a pipelined datapath, the control signals also need to be pipelined systematically.



Assuming, that we decide to place our control unit in the instruction decode stage, we must keep execute (EX), memory (MEM) and write-back (WB) stage control signals in the pipeline register between instruction decode and execute stage (ID/EX).

As each instruction passes through every stage, one cycle at a time, it is important to ensure that their corresponding control signals pass through every stage synchronously, to ensure cycle accuracy.

Lab Task
Design and integrate a pipeline register into the data and control path designed previously.